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Powder sprayer

Abstract

A powder sprayer includes a powder storage tank, a nozzle communicated with the powder storage tank, a telescopic rod, and a balloon. The telescopic rod is communicated with the powder storage tank and is spaced apart from the nozzle. A length of the telescopic rod is adjustable. The balloon is communicated with one end of the telescopic rod away from the powder storage tank. By such configurations, an application range of the powder sprayer is increased and safety of the powder sprayer is improved.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to a field of powder spraying tools, and in particular to a powder sprayer.
BACKGROUND
(2) In the prior art, powder sprayers are widely used in various occasions, such as agricultural pesticide application, industrial spraying, etc. In the prior art, the powder sprayers generally comprise a balloon, a powder storage tank, and a nozzle. Powder is sprayed from the nozzle, and a distance between the nozzle and an operator is relatively close, which makes it easy for the operator to come into close contact with sprayed powder during use. The sprayed powder is easily inhaled by the operator, posing a potential threat to health of the operator.
SUMMARY
(3) The embodiment of the present disclosure provides a powder sprayer, which increases a distance between a nozzle thereof and an operator, reduces a risk of powder being inhaled by the operator, and increases safety.
(4) In a first aspect, the present disclosure provides a powder sprayer. The powder sprayer comprises a powder storage tank, a nozzle communicated with the powder storage tank, a telescopic rod, and a balloon.
(5) The telescopic rod is communicated with the powder storage tank and is spaced apart from the nozzle. A length of the telescopic rod is adjustable. The balloon is communicated with one end of the telescopic rod away from the powder storage tank.
(6) Optionally, the telescopic rod comprises rod portions. The rod portions are sequentially connected between the balloon and the powder storage tank. The rod portions are communicated with each other. Two adjacent rod portions are sleeved with each other and are slidable relative to each other.
(7) Optionally, the two adjacent rod portions are hermetically connected.
(8) Optionally, the rod portions comprise a first rod portion and a second rod portion disposed

adjacent to the first rod portion. The first rod portion is sleeved on an outer side of the second rod portion. The second rod portion is slidable relative to the first rod portion. A thread is disposed on an outer periphery of an end portion of the first rod portion close to the second rod portion.
(9) The telescopic rod further comprises a screw joint. The screw joint is sleeved on the second rod portion and is screwed with the thread of the first rod portion. The screw joint is configured to fix the second rod portion relative to the first rod portion.
(10) Optionally, the telescopic rod further comprises a sealing ring. The sealing ring is sleeved on the second rod portion and disposed between the first rod portion and the second rod portion.
(11) Optionally, a length of at least one of the rod portions is greater than a length of the nozzle.
(12) Optionally, the powder sprayer further comprises a handle. The handle is connected to the one end of the telescopic rod away from the powder storage tank, and the balloon is mounted on the handle.
(13) Optionally, the handle comprises a first end portion and a second end portion disposed opposite to the first portion. The balloon is mounted between the first end portion and the second end portion of the handle. The first end portion of the handle is connected to the telescopic rod.
(14) Optionally, the first end portion of the handle defines an air channel, and the balloon is communicated with the telescopic rod through the air channel.
(15) Optionally, the powder sprayer further comprises a check valve, and the check valve is disposed in the air channel.
(16) Optionally, the handle further comprises a first connecting portion. The first connecting portion connects the first end portion of the handle to the second end portion of the handle. One side of the first connecting portion facing the balloon is arc-shaped and is matched with the balloon.
(17) Optionally, the handle further comprises a second connecting portion disposed opposite to the first connecting portion. The second connecting portion is connected between the first end portion of the handle and the second end portion of the handle. The second connecting portion is spaced apart from the balloon.
(18) Optionally, the powder storage tank comprises a tank body and a cover body. The cover body covers the tank body. The cover body comprises an air inlet and an air outlet. The air inlet is communicated with the telescopic rod. The air outlet is communicated with the nozzle.
(19) Optionally, the powder storage tank further comprises an air inlet pipe and an air outlet pipe. The air inlet pipe is communicated with the air inlet. The air outlet pipe is communicated with the air outlet.
(20) The telescopic rod, the powder storage tank, and the nozzle are disposed in sequence. The air inlet pipe is bent in a direction away from the tank body and towards the telescopic rod. The air outlet pipe is bent in the direction away from the tank body and towards the nozzle.
(21) Optionally, the air inlet pipe protrudes from one side of the cover body toward the tank body.
(22) Optionally, the telescopic rod is screwed with the air inlet pipe, and the nozzle is screwed with the air outlet pipe.
(23) Optionally, the air inlet pipe, the air outlet pipe, and the cover body are integrally formed.
(24) Optionally, the powder sprayer further comprises a check valve, the check valve is disposed between the balloon and the telescopic rod.
(25) Optionally, the powder sprayer further comprises a check valve, the check valve is disposed between the telescopic rod and the powder storage tank.
(26) In the powder sprayer of the embodiment of the present disclosure, the telescopic rod with adjustable length is connected between the balloon and the powder storage tank, so as to increase the distance between the nozzle and the operator by adjusting the length of the telescopic rod, thereby reducing the risk of powder being inhaled and ensuring the safety and health of the operator. Moreover, since there is no need to increase the length of the nozzle, a problem of powder clogging in the nozzle is reduced. In addition, since the length of the telescopic rod is adjustable,

the operator only needs to simply adjust the length of the telescopic rod without frequently changing an operating position or changing tools, so as to adapt to different operating conditions. Therefore, an application range of the powder sprayer is expanded, a convenience of operation and work efficiency is improved, and adaptability and practicality of the powder sprayer are enhanced. When the powder sprayer is not in use, the telescopic rod is allowed to be retracted to a shortest length, making the powder sprayer compact and easy to store and carry.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In order to clearly describe technical solutions in the embodiments of the present disclosure, the following will briefly introduce the drawings that need to be used in the description of the embodiments or the prior art. Apparently, the drawings in the following description are merely some of the embodiments of the present disclosure, and those skilled in the art are able to obtain other drawings according to the drawings without contributing any inventive labor.

(2) For a complete understanding of the present disclosure and its characteristics, the following description will be made in conjunction with the accompanying drawings, where same reference numbers in the following description indicate same structures.

(3) FIG. **1** is a schematic diagram of a powder sprayer according to one embodiment of the present disclosure.

(4) FIG. **2** is another schematic diagram of the powder sprayer according to one embodiment of the present disclosure.

(5) FIG. **3** is a cross-sectional schematic diagram of the powder sprayer taken along a line A-A shown in FIG. **2**.

(6) FIG. **4** is an enlarged schematic diagram of portion B of the powder sprayer shown in FIG. **3**.

(7) FIG. **5** is an enlarged schematic diagram of portion C of the powder sprayer shown in FIG. **3**.

(8) FIG. **6** is an enlarged schematic diagram of portion A of the powder sprayer shown in FIG. **3**.

(9) FIG. **7** is a schematic diagram of a check valve of the powder sprayer shown in FIG. **4**.

DETAILED DESCRIPTION

(10) Technical solutions in the embodiments of the present disclosure will be clearly and completely described below in conjunction with the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments of the present disclosure, rather than all of the embodiments. The following description of at least one exemplary embodiment is merely illustrative and is in no way intended as any limitation on the present disclosure, applications thereof, and use thereof. Based on the embodiments of the present disclosure, all other embodiments obtained by those of ordinary skill in the art without creative work shall fall within the protection scope of the present disclosure.

(11) Reference herein to “embodiment” or “implement” means that a particular feature, structure, or characteristic described in connection with one embodiment or one implement may be comprised in at least one embodiment of the present disclosure. The appearances of the “embodiment” in various positions in the specification are not necessarily referring to the same embodiment, and are not independent or alternative embodiments mutually exclusive of other embodiments. Those skilled in the art explicitly and implicitly understand that the embodiments described herein may be combined with other embodiments.

(12) The present disclosure provides a powder sprayer. As shown in FIG. **1**, FIG. **1** is a schematic diagram of the powder sprayer according to one embodiment of the present disclosure. The powder sprayer comprises a powder storage tank **100**, a nozzle **200**, a telescopic rod **300**, and a balloon **400**. The nozzle **200** is communicated with the powder storage tank **100**. The telescopic rod **300** is communicated with the powder storage tank **100** and is spaced apart from the nozzle **200**. A length

of the telescopic rod **300** is adjustable. The balloon **400** is communicated with one end of the telescopic rod **300** away from the powder storage tank **100**.
(13) The powder storage tank **100** is configured to store powder. The powder may be of various types, which is selected according to specific usage requirements. The nozzle **200** is connected to the powder storage tank **100**. The nozzle **200** is an outlet for spraying powder. Through the nozzle **200**, the powder is sprayed and covered to ae target spraying area. The telescopic rod **300** is connected to the powder storage tank **100**, and the one end of the telescopic rod **300** away from the powder storage tank **100** is communicated with the balloon **400**, so the telescopic rod **300** serves as a transmission device connecting the balloon **400** and the powder storage tank **100**. When the balloon **400** is squeezed, air is squeezed into the telescopic rod **300**, thereby pushing the powder in the powder storage tank **100** to be sprayed out through the nozzle **200**.
(14) In the embodiment of the present disclosure, the powder sprayer **10** comprises the telescopic rod **300** connected between the balloon **400** and the powder storage tank **100**, which obviously increases a distance between the nozzle **200** and an operator. Furthermore, a length of the telescopic rod **300** is adjustable, and the operator is able to adjust an overall length of the powder sprayer **10** by adjusting the length of the telescopic rod **300** according to actual needs. When the telescopic rod **300** is extended longer, the distance between the nozzle **200** and a face of the operator is longer, thereby physically reducing a possibility of the operator inhaling powder, increasing an application range of the powder sprayer and improving operational safety.
(15) It is understood that if there is no telescopic rod **300**, in order to increase the distance between the operator and the nozzle **200**, a length of the nozzle **200** needs to be increased, and an increase in the length of the nozzle **2001** makes a powder conveying path thereof being too long, which increases the risk of powder deposition and clogging. As a compromise, the overall length of the powder sprayer **10** is limited. In the embodiment of the present disclosure, the configuration of the telescopic rod **300** avoids a problem of the nozzle **200** being too long, there is no need to increase the length of the nozzle **200**, and the powder conveying path is no needed to be increased, which reduces the problem of powder clogging.
(16) Moreover, in the embodiment of the present disclosure, the length of the telescopic rod **300** is adjustable, so that the powder sprayer **10** is able to adapt to a variety of usage scenarios and distance requirements. For example, in usage scenarios such as agricultural spraying and industrial spraying, the operator is able to adjust the length of the telescopic rod **300** according to a height and a distance of the target spraying area to accurately control a spraying direction and a distance of the powder, which optimizes a powder spraying effect and ensures that the powder evenly and completely covers the target spraying area.
(17) In the powder sprayer of the embodiment of the present disclosure, the telescopic rod **300** with adjustable length is connected between the balloon **400** and the powder storage tank **100**, so as to increase the distance between the nozzle **200** and the operator by adjusting the length of the telescopic rod **300**, thereby reducing the risk of powder being inhaled and ensuring the safety and health of the operator. Moreover, since there is no need to increase the length of the nozzle **200**, the risk of powder clogging in the nozzle **200** is reduced. In addition, since the length of the telescopic rod **300** is adjustable, the operator only needs to simply adjust the length of the telescopic rod **300** without frequently changing an operating position or changing tools, so as to adapt to different operating conditions. Therefore, the application range of the powder sprayer is expanded, a convenience of operation and work efficiency is improved, and adaptability and practicality of the powder sprayer are enhanced. When the powder sprayer is not in use, the telescopic rod **300** is allowed to be retracted to a shortest length, making the powder sprayer compact and easy to store and carry.
(18) In some embodiments, the telescopic rod **300** comprises rod portions **310**. The rod portions **310** are sequentially connected between the balloon **400** and the powder storage tank **100**. The rod portions **310** are communicated with each other. Two adjacent rod portions **310** are sleeved with

each other and are slidable relative to each other.

(19) In the embodiment, the rod portions **310** are sequentially connected between the balloon **400** and the powder storage tank **100**, so as to from a continuous and unobstructed air transmission channel, ensuring that the air is transferred from the balloon **400** to the powder storage tank **100** through the telescopic rod **300**. The two adjacent rod portions **310** are sleeved with each other and are slidable relative to each other, so that the length of the telescopic rod **300** is flexibly adjusted as needed to adapt to different usage scenarios and different spraying distances. In a specific use process, the operator only needs to adjust the length of the telescopic rod **300** by simply sliding any one of the rod portions **310** according to work requirements, without relying on complex mechanical devices or tools, which enhances the convenience of operation.

(20) In some embodiments, the two adjacent rod portions **310** are hermetically connected. The two adjacent rod portions **310** are hermetically connected. A sealed connection between the two adjacent rod portions **310** ensures that the air does not leak from a joint of the telescopic rod **300** during a compression process or a releasement process of the balloon **400**, thereby ensuring the spraying efficiency of the powder sprayer **10**. During a powder spraying process, the sealed connection between the two adjacent rod portions **310** further prevents the powder from leaking out from the joint of the telescopic rod **300**, keeps a working environment clean, reduces a waste of powder, and protects the health of the operator. The sealed connection between the two adjacent rod portions **310** further reduces an erosion of dust and moisture on an interior of the telescopic rod **300**, thereby extending service life of the powder sprayer **10**. In addition, the sealed connection between the two adjacent rod portions **310** further makes the telescopic rod **300** well adapt to a variety of working environments. Whether in a dry, humid or dusty environment, the sealed connection between the two adjacent rod portions **310** ensures a normal operation of the telescopic rod **300** without being affected by an external environment. Optionally, the sealed connection between the two adjacent rod portions **310** is realized by a sealing ring **350**, an O-ring, a gasket or other sealing elements.

(21) As shown in FIGS. **2-4**, FIG. **2** is another schematic diagram of the powder sprayer according to one embodiment of the present disclosure, FIG. **3** is a cross-sectional schematic diagram of the powder sprayer taken along a line A-A shown in FIG. **2**, and FIG. **4** is an enlarged schematic diagram of portion B of the powder sprayer shown in FIG. **3**. In some embodiments, the rod portions **310** comprise a first rod portion **311** and a second rod portion **312** disposed adjacent to the first rod portion **311**. The first rod portion **311** is sleeved on an outer side of the second rod portion **312**. The second rod portion **312** is slidable relative to the first rod portion **311**. A thread is disposed on an outer periphery of an end portion of the first rod portion **311** close to the second rod portion **312**. The telescopic rod **300** further comprises a screw joint **330**. The screw joint **330** is sleeved on the second rod portion **312** and is screwed with the thread of the first rod portion **311**. The screw joint **330** is configured to fix the second rod portion **312** relative to the first rod portion **311**.

(22) In the embodiment, the first rod portion **311** and the second rod portion **312** are two adjacent rod portions **310** in the telescopic rod **300**. The first rod portion **311** is sleeved on the outside of the second rod portion **312**, and the second rod portion **312** is slidable relative to the first rod portion **311**. Whether the length of the telescopic rod **300** needs to be extended or shortened, the operator only needs to apply appropriate force to push or pull the second rod portion **312** to slide in the first rod portion **311**, and the length of the telescopic rod **300** is quickly and easily adjusted to adapt to different working scenarios and spraying distances without the need for complicated operating steps or tool assistance.

(23) Moreover, in the embodiment, the thread is disposed on the outer periphery of the end portion of the first rod portion **311** close to the second rod portion **312**, and the telescopic rod **300** further comprises the screw joint **330** that is sleeved on the second rod portion **312** and screwed with the thread of the first rod portion **311**. After the length of the telescopic rod **300** is adjusted by the operator, the second rod portion **312** and the first rod portion **311** is fixed by rotating the screw joint

330, thereby ensuring that the telescopic rod **300** maintains a stable length during operation, enhancing the stability of a structure of the telescopic rod **300**, effectively avoiding a risk of failure caused by relative sliding between the rod portions **310** during operation, and improving working efficiency and reliability of the powder sprayer **10**. When the length of the telescopic rod **300** needs to be adjusted again, the operator only needs to rotate the screw connection **330** to unlock it, so there is no need to use complex mechanical devices or additional tools to fix and unlock the telescopic rod **300**, thereby improving the operating efficiency.

(24) In some embodiments, the telescopic rod **300** further comprises a sealing ring **350**. The sealing ring **350** is sleeved on the second rod portion **312** and disposed between the first rod portion **311** and the second rod portion **312**. The screw joint **130** is screwed with the first rod portion **311** and enables the first rod portion **311** and the second rod portion **312** to compress the sealing ring **350**.

(25) In the embodiment, the sealing ring **350** is disposed between the first rod portion **311** and the second rod portion **312**. The sealing ring **350** effectively fills a gap defined between the first rod portion **311** and the second rod portion **312**, thereby preventing the air from leaking out of the telescopic rod **300** and ensuring that the air is able to sufficiently and forcefully push the powder in the powder storage tank **100** to be sprayed out from the nozzle **200**. Thus, powder spraying efficiency is improved, and the powder is effectively prevented from leaking out of the telescopic rod **300**, which helps to protect the health of operator and reduce an impact on the external environment, so that the operator is able to use the powder sprayer **10** assuredly. In addition, by providing the sealing ring **350**, the interior of the telescopic rod **300** is protected from the influence of the external environment, such as dust, moisture, etc., thereby extending the service life of the telescopic rod **300**.

(26) In some embodiments, a length of at least one of the rod portions **310** is greater than the length of the nozzle **200**. At least one of the rod portions **310** is greater than the length of the nozzle **200**, so the nozzle **200** of the powder sprayer **10** is kept at a relatively short length. In the embodiment, it is not necessary to increase the distance between the operator and the sprayed powder by increasing the length of the nozzle **200**. Instead, the rod portions **310** that are relatively long are provided, so that the operator is able to control the powder sprayer **10** to accurately spray powder at a farther distance, effectively increasing a physical distance between the operator and the target spraying are, and increasing the safety of the operation. In addition, the operator is allowed to stand in a relatively fixed position, and by adjusting a placing angle of the powder sprayer **10** and an extension direction of the rod portions **310**, it is easier to complete a large-area powder spraying operation without frequently movement, making the operation convenient and efficient.

(27) As shown in FIG. **5**, FIG. **5** is an enlarged schematic diagram of portion C of the powder sprayer shown in FIG. **3**. In some embodiments, the powder sprayer further comprises a handle **700**. The handle **700** is connected to the one end of the telescopic rod **300** away from the powder storage tank **100**, and the balloon **400** is mounted on the handle **700**. Configuration of the handle **700** allows the operator to hold the powder sprayer **10** comfortably. A design of the handle **700** takes into account both ergonomics and practicality, so the operator is allowed to complete the powder spraying operation in a relaxed and natural way. The balloon **400** is mounted on the handle **700**, so that the operator is enable to naturally touch the balloon **400** when holding the handle **700**, and the spraying of the powder is easily controlled.

(28) In some embodiments, the handle **700** comprises a first end portion **750** and a second end portion **760** disposed opposite to the first portion. The balloon **400** is mounted between the first end portion **750** and the second end portion **760** of the handle **700**. The first end portion **750** of the handle **700** is connected to the telescopic rod **300**. The balloon **400** is mounted between the first end portion **750** and the second end portion **760** of the handle **700**, so the balloon **400** and the handle **700** form a compact whole, which provides stable support, reduces shaking of the balloon **400** during use, and improves accuracy of spraying powder. The first end portion **750** is directly connected to the telescopic rod **300**, which ensures that the force applied by the operator to the

balloon **400** is directly and effectively transmitted to the telescopic rod **300**, thereby improving the powder spraying efficiency of the powder sprayer **10**.

(29) In some embodiments, the first end portion **750** of the handle **700** defines an air channel **711**, and the balloon **400** is communicated with the telescopic rod **300** through the air channel **711**. The air channel **711** serves as a medium connecting the balloon **400** and the telescopic rod **300**, ensuring that the air input by squeezing the balloon **400** is smoothly transmitted to the interior of the telescopic rod **300**, reducing air leakage during the transmission process, ensuring that more air is transmitted to spray the powder, and improving performance of the powder sprayer **10**.

(30) In some embodiments, the powder sprayer further comprises a check valve **800**, and the check valve **800** is disposed in the air channel **711**. The check valve **800** is disposed in the air channel **711**, so the check valve **800** is well protected from interference and damage from the external environment, while ensuring one-way flow of the air. Therefore, the powder in the powder storage tank **100** is prevented from being brought back into the telescopic rod **300** or the balloon **400** by the air during the spraying process.

(31) In some embodiments, the handle **700** further comprises a first connecting portion **730**. The first connecting portion **730** connects the first end portion **750** of the handle **700** to the second end portion **760** of the handle **700**. One side of the first connecting portion **730** facing the balloon **400** is arc-shaped and is matched with the balloon **400**. The first connecting portion **730** connects the first end portion **750** of the handle **700** to the second end portion **760** of the handle **700**, and the one side of the first connecting portion **730** facing the balloon **400** is arc-shaped and is matched with the balloon **400**. An arc-shaped design of the one side of the first connecting portion **730** enables the first connecting portion **730** to better fit a shape of the balloon **400**, thereby optimizing a spatial layout between the handle **700** and the balloon **400**.

(32) A shape of the first connection part **730** is an arc shape that matches the balloon **400**. Therefore, the operator can easily hold the first connection part **730** with one hand and naturally apply pressure to the balloon **400** with four fingers. When the fingers squeeze the balloon **400**, the air inside the balloon **400** is pushed and quickly blown into the powder storage tank **100** through the telescopic rod **300** connected thereto. In the process, the powder in the powder storage tank **100** is driven by the strong airflow and sprayed out from the nozzle **200**, realizing an efficient powder spraying function.

(33) In some embodiments, the handle **700** further comprises a second connecting portion **740** disposed opposite to the first connecting portion **730**. The second connecting portion **740** is connected between the first end portion **750** of the handle **700** and the second end portion **760** of the handle **700**. The second connecting portion **740** is spaced apart from the balloon **400**. The second connecting portion **740** is also connected between the first end portion **750** of the handle **700** and the second end portion **760** of the handle **700**. The second connection portion **740** and the first connection portion **730** form a ring structure. The ring structure surrounds the balloon **400** in a middle portion thereof, so the balloon **400** is better protected and a risk of damage to the balloon **400** due to accidental collision or squeezing during use is reduced. A certain distance is defined between the second connection portion **740** and the balloon **400** to ensure that the second connection portion **740** does not affect a squeezing operation of the operator on the balloon **400**.

(34) By configurations of the first connecting portion **730** and the second connecting portion **740**, an overall structure of the handle **700** is made stable, which not only improves the reliability of the powder sprayer **10** during operation, but also enables the handle **700** to adapt to hand shapes and holding habits of different operators. The operator is able to choose different holding positions on the ring structure according to his/her preference and work requirements, thereby obtaining a comfortable holding experience.

(35) As shown in FIG. **6**, FIG. **6** is an enlarged schematic diagram of portion A of the powder sprayer shown in FIG. **3**. In some embodiments, the powder storage tank **100** comprises a tank body **110** and a cover body **120**. The cover body **120** covers the tank body **110**. The cover body **120**

comprises an air inlet **121** and an air outlet **122**. The air inlet **121** is communicated with the telescopic rod **300**. The air outlet **122** is communicated with the nozzle **200**.

(36) In the embodiment, the tank body **110** of the powder storage tank **100** is configured to store the powder, and the cover body **120** is integrated with the air inlet **121** and the air outlet **122**. The air inlet **121** is communicated with the telescopic rod **300**, and the air outlet **122** is communicated with the nozzle **200**, which ensures that the air is able to smoothly slowing into and flowing out of the powder storage tank **100** and then effectively drive the powder to be sprayed through the nozzle **200**. The air inlet **121** and the air outlet **122** are defined in the cover body **120** of the powder storage tank **100** instead of the tank body **110**, so that when the air is in a static state, the powder is not accumulated in the air inlet **121** and the air outlet **122**, which effectively avoids a clogging problem caused by powder accumulation and ensures smooth flow of the air in the air inlet **121** and the air outlet **122**. Therefore, in the powder spraying operation, the air quickly and smoothly enters the tank body **110** through the air inlet **121** and drives the powder to be efficiently sprayed out from the air outlet **122**. In addition, the air inlet **121** and the air outlet **122** are defined in the cover body **120**, so that connections between the powder storage tank **100**, the telescopic rod **300**, and the nozzle **200** are flexible and convenient. When it is necessary to replace or add the powder, it is only necessary to open the cover body **120** to easily move the tank body **110** for operation, which brings a convenient and efficient use experience to the operator.

(37) In some embodiments, the powder storage tank **100** further comprises an air inlet pipe **123** and an air outlet pipe **124**. The air inlet pipe **123** is communicated with the air inlet **121**. The air outlet pipe **124** is communicated with the air outlet **122**. The telescopic rod **300**, the powder storage tank **100**, and the nozzle **200** are disposed in sequence. The air inlet pipe **123** is bent in a direction away from the tank body **110** and towards the telescopic rod **300**. The air outlet pipe **124** is bent in the direction away from the tank body **110** and towards the nozzle **200**.

(38) In the embodiment, the air inlet pipe **123** that is bent directly and efficiently guides the air input by squeezing the balloon **400** from the telescopic rod **300** to the tank body **110**. The air mixed with the powder in the tank body **110** is guided from the air outlet pipe **124** that is bent to the nozzle **200**, thereby enhancing a guiding effect of the air. Thus, the air is allowed to flow smoothly along a predetermined path, so as to reduce air flow turbulence and energy loss, and improve the powder conveying efficiency and spraying quality.

(39) In some embodiments, the air inlet pipe **123** protrudes from one side of the cover body **120** toward the tank body **110**. The air inlet pipe **123** protrudes from the cover body **120** and towards the tank body **110**, that is, the air inlet pipe **123** is inserted into the tank body **110**, and the air inlet pipe **123** is close to or deep into a powder storage area, and the air from the air inlet pipe **123** directly acts on more powder. When the air enters through the air inlet pipe **123**, it is fully mixed with the powder, and the powder is smoothly carried by the air and is transferred to the nozzle **200**. Further, the air is able to blow the powder in the tank body **110** during a flow process, so the accumulation of the powder on walls or corners of the tank body **110** is reduced, thereby reducing powder residue and improving powder discharge efficiency and a powder utilization rate.

(40) In some embodiments, the telescopic rod **300** is screwed with the air inlet pipe **123**, and the nozzle **200** is screwed with the air outlet pipe **124**. The telescopic rod **300** and the air inlet pipe **123**, as well as the nozzle **200** and the air outlet pipe **124**, are connected by screw connection. During the operation of the powder sprayer **10**, the screw connection thereof withstands greater pressure and vibration, ensuring that the telescopic rod **300**, the air inlet pipe **123**, the nozzle **200**, and the air outlet pipe **124** are tightly and stably connected, and are not prone to loosening or falling off. Moreover, the screw connection is convenient for disassembly. When one of components needs to be repaired, cleaned, or replaced, the one of the components is easily separated by simply rotating the one of the component in an opposite direction, thereby reducing maintenance costs. In addition, as a common mechanical connection method, the screw connection is easy and quick to operate, does not require complex tools or skills, and improves mounting

efficiency.
(41) Since the telescopic rod **300** is screwed with the air inlet pipe **123**, the telescopic rod **300** is rotatable relative to the powder storage tank **100**. The operator is able to adjust a direction of the telescopic rod **300** relative to the powder storage tank **100** as needed. By changing the direction of the telescopic rod **300**, an airflow direction of the air in the air inlet **121** is indirectly adjusted, thereby changing a direct direction of the airflow on the powder. A driving force of the airflow on the powder may change as the direction of the airflow changes, thereby affecting a flow rate and a powder output of the powder. Different spraying operations may require different powder outputs. By adjusting the direction of the telescopic rod **300**, the operator is able to accurately control the powder output to meet different spraying requirements, thereby enhancing the user experience.
(42) In some embodiments, the air inlet pipe **123**, the air outlet pipe **124**, and the cover body **120** are integrally formed. The air inlet pipe **123**, the air outlet pipe **124** and the cover body **120** are designed to be integrally formed. The air inlet pipe **123**, the air outlet pipe **124**, and the cover body **120** are integrated into an integral component during the manufacturing process, which reduces the number of structural parts. There is no need for subsequent assembly steps between the air inlet pipe **123**, the air outlet pipe **124** and the cover body **120**, which reduces a difficulty of operation during assembly, debugging and use for the operator and improves the overall convenience of use. In addition, an integrally formed design thereof enhances a connection strength between the air inlet pipe **123**, the air outlet pipe **124**, and the cover body **120**, reduces a risk of failure caused by loose or falling parts, and improves structural stability and reliability. Of course, in some other embodiments, the air inlet pipe **123**, the air outlet pipe **124**, and the cover body **120** are separated components.
(43) Optionally, the handle **700** and the telescopic rod **300** are made of a lightweight and hard material, which helps to reduce a weight of the powder sprayer **10** and make the powder sprayer **10** easy to operate. The handle **700** and the telescopic rod **300** are hard, so the handle **700** and the telescopic rod **300** are durable and are able to withstand pressure and wear in daily operation, thereby making the overall structure of the powder sprayer **10** stable. The operator is allowed to easily spray the powder on the target spraying area even with one hand. Whether it is in a narrow space or needs to spray powder at a high place, the powder sprayer can easily cope with it. In addition, in the embodiment, the telescopic rod **300** and the handle **700** are hard, which is also an important factor in the powder sprayer **10** being able to achieve one-handed operation. If the telescopic rod **300** and the handle **700** are not hard, and the balloon **400** is soft, the handle **700** is unable to provide enough structural support for the balloon **400**, and the telescopic rod **300** is unable to bear a weight of the powder storage tank **100** and the nozzle **200**. As a result, the operator must hold the balloon **400** in one hand to operate, and hold the powder storage tank **100** in the other hand, which causes inconvenience in operation. In contrast, since the telescopic rod **300** and the handle **700** are hard, the overall structure of the powder sprayer **10** is stable and reliable. The telescopic rod **300** that is hard is able to transmit the force applied by the operator to ensure that the powder is sprayed evenly and powerfully. The handle **700** that is hard provides a good grip feel and stability, so that the operator is able to easily control the powder sprayer **10** with one hand to point to the target spraying area that needs to be sprayed. Therefore, the configurations of the telescopic rod **300** and the handle **700** that are hard in the embodiment improve the convenience and flexibility of operation of the powder sprayer **10**.
(44) As shown in FIG. **7**, FIG. **7** is a schematic diagram of the check valve of the powder sprayer shown in FIG. **4**. In the embodiments, the powder sprayer further comprises a check valve **800**. The check valve **800** is configured to control one-way flow of the air, ensure that the air flows smoothly on a specific path, and prevent the air from flowing back. Optionally, one or more check valves **800** are provided. When check valves are provided, the check valves **800** are flexibly disposed at a plurality of positions to ensure a stable operation of the powder sprayer **10**, protect the safety of the operator, and improve the powder spraying effect.

(45) In some embodiment, when only one check valve **800** is provided, the check valve **800** is disposed between the balloon **400** and the telescopic rod **300**. When the operator squeezes the balloon **400**, the air in the balloon **400** is pushed and flows to the telescopic rod **300** through the check valve **800**, then the air pushes the powder in the powder storage tank **100** to be sprayed out. After the operator releases the balloon **400**, the balloon **400** restores to an original shape and form a negative pressure area in a restore process. The check valve **800** effectively prevents the air in the telescope rod **300** from flowing back to the balloon **400**. If there is no check valve **800**, the air in the powder storage tank **100** may flow back to the telescopic rod **300** and the balloon **400** under an action of negative pressure, which reduces the powder spraying effect. More importantly, back flowing air is mixed with the powder, which increases the risk of the powder being physically in contact with the operator. In the embodiment, by providing the check valve **800**, when squeezing the balloon **400**, the air is allowed to flow out smoothly; and when the balloon **400** restores the original shape, the air is prevented from flowing back, thereby avoiding the risk of the air and the powder in the powder storage tank **100** being inhaled. Therefore, the performance of the powder sprayer **10** is improved and the health and safety of the operator are ensured. For example, the balloon **400** restores the original shape after being released, and fresh air is sucked from one end of the balloon **400** away from the powder storage tank **100**, rather than sucking the air and the powder in the powder storage tank **100**.
(46) In some embodiments, the powder sprayer **10** comprises only one check valve **800**. The check valve **800** is disposed between the telescopic rod **300** and the powder storage tank **100**, which ensures that the air is only allowed to flow from the telescopic rod **300** to the powder storage tank **100**, but is unable to flow back from the powder storage tank **100** to the telescopic rod **300**, thereby ensuring that the powder in the powder storage tank **100** is not brought back to the telescopic rod **300** or the balloon **400** by the air during the spraying process, ensuring the health and safety of the operator, and helping to maintain the stability and reliability of the powder sprayer **10**.
(47) In some alternative embodiments, the powder sprayer **10** comprises only one check valve **800**. The check valve is disposed in the telescopic rod **300**. The check valve **800** is disposed inside, so the check valve **800** is prevented from interference and damage from the external environment. Further, the check valve **800** also ensures the one-way flow of the air and prevents the powder in the powder storage tank **100** from being brought back into the telescopic rod **300** or the balloon **400** by the air during the spraying process.
(48) The most suitable configuration method of the one or more check valves **800** is selected according to specific application scenario and requirements. There may be one check valve, or the check valves **800** are respectively disposed between the telescopic rod **300** and the powder storage tank **100**, inside the telescopic rod **300**, and between the telescopic rod **300** and the balloon **400**, which is not limited thereto.
(49) In the foregoing embodiments, the description of each of the embodiments has its own emphasis, and some features that are not detailed or described in some of the embodiments may refer to related descriptions of other embodiments. The embodiments, implementations and related technical features of the present disclosure can be combined and replaced with each other without conflict.
(50) The above embodiments of the present disclosure provide a detailed illustration to the powder sprayer. In the present disclosure, specific embodiments are applied to illustrate the principles and implementations of the present disclosure. The above description of the embodiments is only used to better understand methods and core ideas of the present disclosure. Meanwhile, according to the ideas of the present disclosure, changes are made in the specific implementations and the application scope by those skilled in the art. Therefore, the contents of the specification should not be regarded as a limitation of the present disclosure.

Claims

1. A powder sprayer, comprising: a powder storage tank; a nozzle communicated with the powder storage tank; a telescopic rod; and a balloon; wherein the telescopic rod is communicated with the powder storage tank and is spaced apart from the nozzle, a length of the telescopic rod is adjustable, and the balloon is communicated with one end of the telescopic rod away from the powder storage tank; wherein the powder sprayer further comprises a handle, the handle is connected to the one end of the telescopic rod away from the powder storage tank, and the balloon is mounted on the handle; wherein the handle comprises a first end portion and a second end portion disposed opposite to the first portion, the balloon is mounted between the first end portion and the second end portion of the handle, and the first end portion of the handle is connected to the telescopic rod; wherein the handle further comprises a first connecting portion, the first connecting portion connects the first end portion of the handle to the second end portion of the handle, and wherein one side of the first connecting portion facing the balloon is arc-shaped and is matched with the balloon; wherein the handle further comprises a second connecting portion disposed opposite to the first connecting portion, the second connecting portion is connected between the first end portion of the handle and the second end portion of the handle, and the second connecting portion is spaced apart from the balloon, wherein the second connection portion and the first connection portion form a closed ring structure that surrounds the balloon in a middle region thereof.
2. The powder sprayer according to claim 1, wherein the telescopic rod comprises a plurality of rod portions, the plurality of rod portions are sequentially connected between the balloon and the powder storage tank, the plurality of rod portions are communicated with each other, and wherein every two adjacent rod portions are sleeved with each other and are slidable relative to each other.
3. The powder sprayer according to claim 2, wherein the two adjacent rod portions are hermetically connected.
4. The powder sprayer according to claim 3, wherein the rod portions comprise a first rod portion and a second rod portion disposed adjacent to the first rod portion, the first rod portion is sleeved on an outer side of the second rod portion, the second rod portion is slidable relative to the first rod portion, and a thread is disposed on an outer periphery of an end portion of the first rod portion close to the second rod portion; wherein the telescopic rod further comprises a screw joint, the screw joint is sleeved on the second rod portion and is screwed with the thread of the first rod portion, and the screw joint is configured to fix the second rod portion relative to the first rod portion.
5. The powder sprayer according to claim 4, wherein the telescopic rod further comprises a sealing ring, and the sealing ring is sleeved on the second rod portion and disposed between the first rod portion and the second rod portion.
6. The powder sprayer according to claim 2, wherein a length of at least one of the rod portions is greater than a length of the nozzle.
7. The powder sprayer according to claim 1, wherein the first end portion of the handle defines an air channel, and the balloon is communicated with the telescopic rod through the air channel.
8. The powder sprayer according to claim 7, wherein the powder sprayer further comprises a check valve, and the check valve is disposed in the air channel.
9. The powder sprayer according to claim 1, wherein the powder storage tank comprises a tank body and a cover body, the cover body covers the tank body, the cover body comprises an air inlet and an air outlet, the air inlet is communicated with the telescopic rod, and the air outlet is communicated with the nozzle.
10. The powder sprayer according to claim 9, wherein the powder storage tank further comprises an air inlet pipe and an air outlet pipe, the air inlet pipe is communicated with the air inlet, and the

air outlet pipe is communicated with the air outlet; wherein the telescopic rod, the powder storage tank, and the nozzle are disposed in sequence, the air inlet pipe is bent in a direction away from the tank body and towards the telescopic rod, and the air outlet pipe is bent in the direction away from the tank body and towards the nozzle.

11. The powder sprayer according to claim 10, wherein the air inlet pipe protrudes from one side of the cover body toward the tank body.

12. The powder sprayer according to claim 10, wherein the telescopic rod is screwed with the air inlet pipe, and the nozzle is screwed with the air outlet pipe.

13. The powder sprayer according to claim 12, wherein the air inlet pipe, the air outlet pipe, and the cover body are integrally formed.

14. The powder sprayer according to claim 1, wherein the powder sprayer further comprises a check valve, and the check valve is disposed between the balloon and the telescopic rod.

15. The powder sprayer according to claim 1, wherein the powder sprayer further comprises a check valve, and the check valve is disposed between the telescopic rod and the powder storage tank.